

Play Mode Camera Sync - User Manual

Unity Editor Tool

Sync your Scene view camera with your game camera during play mode

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1. Introduction

What is Play Mode Camera Sync?

Play Mode Camera Sync is a Unity Editor tool that automatically syncs your Scene view camera with your game camera during play mode. Switch between following your main camera or any GameObject, with easy controls for viewing angle and distance.

Key Benefits

- See exactly what your camera sees in real-time
- Follow any GameObject with one click
- Quick access to standard camera angles
- Adjust viewing distance on the fly
- Zero setup required - works immediately after import
- Remembers your settings across sessions

Core Concepts

Camera Mode - Scene view automatically follows Camera.main, matching its position, rotation, and field of view.

Transform Mode - Scene view locks to a specific GameObject's position with customizable viewing angles.

View Presets - Pre-configured camera angles (Front, Back, Left, Right, Top, Bottom, Free) for Transform mode.

Follow Target - The GameObject currently being tracked in Transform mode.

Design Philosophy

Play Mode Camera Sync is designed to:

- Work automatically with zero configuration
- Stay unobtrusive when disabled
- Provide instant visual feedback
- Persist preferences across sessions

Unity Version Support

- Minimum: Unity 2021.3 LTS
- Recommended: Unity 2022.3 LTS or later
- Tested: 2021 LTS, 2022 LTS, 2023 LTS, 2024+

Compatible with all render pipelines: Built-in, URP, and HDRP.

2. Getting Started

Installation

Unity Package Manager

1. Download the Play Mode Camera Sync package
2. In Unity, go to Assets > Import Package > Custom Package
3. Select the downloaded package
4. Click Import to import all files

Verify Installation

1. Open Unity
2. Check that Tools > Play Mode Camera Sync menu appears
3. The tool is now active - no further setup needed

Quick Start

Step 1: Activate

1. Enter Play Mode (click Play button or press Ctrl/Cmd+P)
2. Go to Tools > Play Mode Camera Sync > Toggle Overlay

3. A small floating window appears

Step 2: Use

1. Scene view now follows your main camera automatically
2. Click the lock icon to toggle on/off
3. Click the camera/transform icon to switch modes

Step 3: Follow Objects

1. Hold Ctrl+Shift and click any object in Scene view, or
2. Click the transform icon and drag an object to the target field
3. Choose a view preset from the dropdown

3. Using the Overlay

Overlay Interface

When you open the overlay, a compact floating window appears showing:

Lock Icon

- Green - Camera mode active
- Blue - Transform mode active
- Gray - Disabled or not in play mode
- Click to toggle sync on/off

Mode Icon

- Camera icon - Following main camera
- Transform icon - Following a GameObject
- Click to switch between modes

Target Field (Transform mode only)

- Shows currently followed GameObject
- Drag and drop objects here
- Clear to return to Camera mode

View Dropdown (Transform mode with target)

- Choose camera angle
- Options: Free, Front, Back, Left, Right, Top, Bottom
- Free mode matches object rotation

Navigation

Mouse

- Click and drag the top bar to move the window
- Click × to close
- Hover over controls to see them highlight

Window Management

- Position is remembered between sessions
- Auto-reopens if it was open when you last closed Unity
- Reset position: Tools > Play Mode Camera Sync > Reset Window Size

4. View Presets

How Presets Work

In Transform mode with a follow target set, view presets let you quickly snap to standard camera angles around the object.

Available Presets

Free

- Scene view matches the object's rotation
- Like Camera mode but follows a transform
- Default mode

Front

- View from front of object
- Looking at object face-on

Back

- View from behind object
- Looking at object from behind

Left

- View from left side of object

Right

- View from right side of object

Top

- View from above object
- Looking straight down

Bottom

- View from below object
- Looking straight up

Switching Presets

- Use the dropdown in the overlay window
- Scene view instantly updates to the new angle
- Distance is preserved when switching

Auto-Switch to Free

- Right-click and drag in Scene view to rotate manually
- Automatically switches to Free mode
- Prevents fighting between your input and the preset

5. Settings

Access settings via: **Tools > Play Mode Camera Sync**

Toggle Overlay

- Show/hide the overlay window
- Window reopens in same position

Distance Control

Mouse Scroll Wheel

- Scroll up to zoom out
- Scroll down to zoom in
- Must be in Play Mode with sync enabled
- Must hover over Scene view

Camera Mode Distance

- Range: 0.1 - 10 units

- Default: 1 unit
- Useful for seeing context around camera

Transform Mode Distance

- Range: 1 - 50 units
- Default: 5 units
- Larger range for following characters/vehicles

Separate Values

- Camera and Transform modes have independent distances
- Each is saved and restored separately

6. Tips & Best Practices

Workflow Integration

Character Controllers

- Use Transform mode with Top view to see movement from above
- Switch to Back view to debug facing direction
- Adjust distance to frame the character optimally

AI Debugging

- Follow AI agents in Transform mode
- Use Free mode to see their facing direction
- Quick-switch between agents with Ctrl+Shift+Click

Camera Testing

- Stay in Camera mode
- Pull back with scroll wheel to see camera shake
- Toggle overlay off to reduce UI clutter

Vehicles

- Follow vehicle in Transform mode
- Top view for steering visualization
- Back view for acceleration/braking

Distance Tips

Best Distances by Use Case

- First-person character: 2-5 units
- Third-person character: 5-15 units
- Vehicle: 10-30 units
- Large objects: 20-50 units
- Camera shake testing: 3-8 units

Keyboard Efficiency

- Keep overlay open during play sessions
- Use Ctrl+Shift+Click for rapid target switching
- Position overlay in a consistent spot

Performance

Play Mode Camera Sync is optimized:

- Zero runtime overhead (editor-only)
- No impact on game performance
- Only updates during play mode when enabled

7. Troubleshooting

Common Issues

Scene View Not Syncing

Symptom: Scene view doesn't follow camera or target

Solutions:

1. Ensure you're in Play Mode
2. Click lock icon to enable (should turn green/blue)
3. Check Camera.main exists in your scene
4. Click in Scene view to focus it

Follow Target Lost

Symptom: Target field shows None after entering play mode

Solutions:

1. Object must exist in scene before play mode
2. Check console for "Could not restore follow target" warning

3. Re-set target with Ctrl+Shift+Click

Overlay Window Disappeared

Symptom: Can't find the overlay window

Solutions:

1. Go to Tools > Play Mode Camera Sync > Toggle Overlay

Distance Not Changing

Symptom: Scroll wheel doesn't adjust distance

Solutions:

1. Ensure you're in Play Mode
2. Enable sync (green/blue lock icon)
3. Mouse must be over Scene view
4. Click in Scene view to focus it

Lock Icon Is Gray

Symptom: Lock icon appears gray

Cause: Not in Play Mode (expected behavior)

Solution: Enter Play Mode for sync to activate

Ctrl+Shift+Click Not Working

Symptom: Holding modifiers and clicking doesn't set target

Solutions:

1. Ensure you're clicking on a visible object
2. Hold both Ctrl and Shift
3. Ensure Scene view is active (not Game view)

Check for visual feedback - hovering with Ctrl+Shift should show "Click to follow: [ObjectName]"

8. Technical Reference

How It Works

Play Mode Camera Sync updates the Scene view camera every frame during play mode to match either:

- Camera.main's position and rotation (Camera mode), or
- A GameObject's position with chosen rotation (Transform mode)

Detection Methods

- Automatically finds Camera.main
- Falls back to first Camera if no main camera
- Supports both perspective and orthographic cameras

Data Storage

Settings stored in EditorPrefs:

- Enable/disable state
- Follow mode (Camera or Transform)
- Camera view distance
- Transform view distance
- Follow target path
- Overlay window position

Settings persist across:

- Unity sessions
- Project switches
- Editor updates

Compatibility

Render Pipelines

- Built-in Render Pipeline: ✓
- Universal Render Pipeline (URP): ✓
- High Definition Render Pipeline (HDRP): ✓

Editor Platforms

- Windows ✓
- macOS ✓
- Linux ✓

Note: Play Mode Camera Sync is editor-only. No runtime components are included or needed.

File Location

The script should be placed in an Editor folder: Assets/Editor/PlayModeCameraSync.cs

Or anywhere with #if UNITY_EDITOR wrapping (included by default).

Credits & Attribution

Play Mode Camera Sync for Unity Version 1.0.0

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Thank you for using Play Mode Camera Sync!